UNIT - IV : POWDER METALLURGY

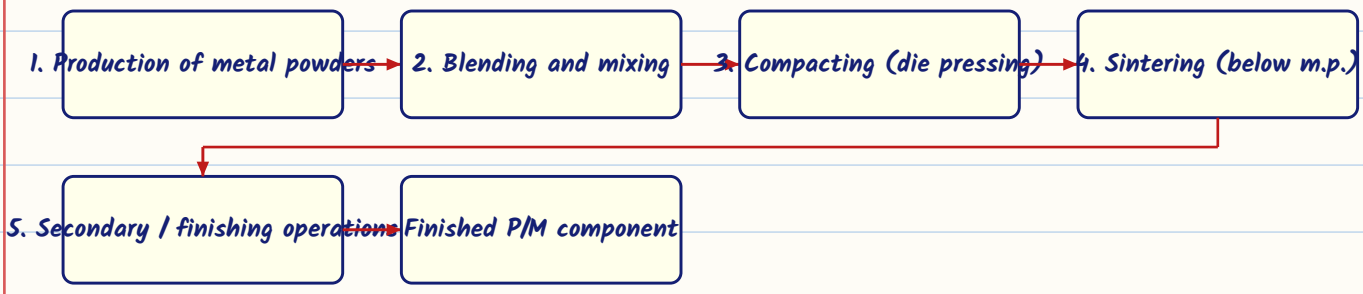
I. Introduction

* POWDER METALLURGY (P/M) = art and science of producing metal powders and making useful engineering components by COMPACTING the powders and SINTERING them (heating below the melting point) - without melting the main constituent.
* History: used by Egyptians (iron tools, 3000 BC), Incas (platinum), 1909 Coolidge made tungsten lamp filaments by P/M; 1930s cemented carbides, porous bearings; today automobile parts form the largest market.

Why powder metallurgy ?

* Metals of VERY HIGH melting point (W 3410 C, Mo, Ta) are difficult to cast; alloys of metals that do not mix in the liquid state (Cu-Pb, W-Cu, Fe-Cu) or metal + non-metal composites (WC-Co, metal-graphite) can be made only by P/M.
* Near net shape parts with close tolerance (+- 0.025 mm) and good surface finish; little or no machining; material utilisation up to 95-97%; controlled porosity (filters, self-lubricating bearings); mass production of small complex parts.

Basic steps of P/M process

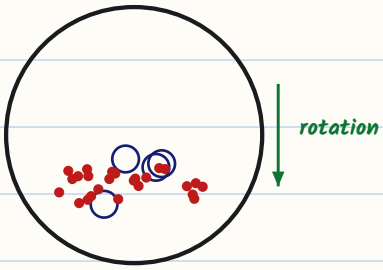


2. Methods of Producing Metal Powders

Properties of powder depend on the method: particle size, shape, purity, apparent density, flow rate, compressibility.

(a) Milling (mechanical pulverisation / crushing)

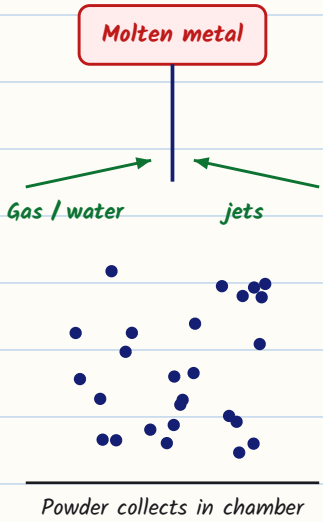
* Brittle metals (Sb, Bi, Cr, Mn, ferro-alloys, oxides, hard metals) and also hard brittle scrap are crushed and ground in BALL MILLS, ROD MILLS, hammer mills, vibratory mills, attritors. Ductile metals get flaky powder (Eddy mill, stamp mill, Hametag process).
* BALL MILL: rotating cylindrical drum partly filled with steel / tungsten carbide balls (30-50% of volume); critical speed N_c = 42.3/sqrt(D) rpm; working speed about 65-75% of N_c. Impact + attrition of balls break the charge. Wet or dry, with inert atmosphere (Ar) for reactive metals. Mechanical alloying (high energy milling) produces ODS alloys.
* Powder: angular, irregular; contamination from the balls; work hardened; cheap but slow process.



Ball mill

Low speed: balls roll / cascade (attrition, fine powder).
Correct speed: balls fall in a cataract - impact breaks charge.
Above Nc balls stick to the wall (centrifuging) - no milling.

(b) Atomization (most widely used - for Al, Fe, steels, Cu, Sn, Zn, P



Atomization

Melt stream broken by high pressure gas / water jets into fine droplets that freeze as powder.
Particle size 10 - 300 micron;
water -> irregular, gas -> spherical.

* Liquid metal stream (from tundish through a nozzle) is disintegrated into droplets by a high pressure jet of water, gas (N2, Ar, air) or by centrifugal force. Droplets solidify quickly into powders.
* Water atomization: irregular shape, cheap, higher oxygen. Gas atomization: spherical, clean, high flowability, costly (super alloys, Ti). Centrifugal / rotating electrode (PREP) : very clean spherical powder. Ultrasonic / vacuum atomization also used. Rapid solidification gives fine microstructure.

(c) Granulation

* Molten metal poured through a screen into water (or stirred while

cooling) - gives coarse granules or shot, used for low melting metals (Zn, Sn, Pb, Cu, Al, Cd). Particles are rough and irregular; used for making pellets, shot and raw material for secondary processes.

(d) Reduction of oxides

* Metal oxide (mill scale, iron ore, tungsten oxide) is reduced in solid state at temperature below m.p. using a reducing gas (H2, CO, cracked ammonia) or carbon: MeO + H2 -> Me + H2O.
* Used for Fe, Cu, W, Mo, Ni, Co. $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$ (Hoganas sponge iron process). Powder is porous, SPONGY (sponge iron) with good compressibility and green strength. Particle size is controlled by oxide size and temperature. Cheap, high production.

(e) Electrolytic deposition

* Metal ions from an electrolyte are deposited at the cathode at high current density as spongy / powdery / brittle deposit, which is scraped, washed, dried, annealed and milled. Cu powder (CuSO4 + H2SO4 bath), Fe, Ni, Zn, Cr, Mn, Ag powders.
* Very pure (99.5-99.9%), dendritic powders with excellent green strength; but high cost and energy.

(f) Other methods

* Chemical: thermal decomposition of carbonyls, e.g. Fe(CO)5 -> Fe + 5 CO (very pure, spherical, micron-sized carbonyl iron and nickel powders); precipitation from solutions (hydro-metallurgy); hydride-dehydride (Ti, Zr); plasma rotating electrode process; Spray deposition.

Method	Powder shape	Metals	Remarks
Milling	Angular / flaky	Brittle metals, oxides, carbides	Cheap, contamination
Atomization	Spherical (gas), irregular (water)	Al, Fe, Cu, Sn, steels, alloys	Most common; alloy powders
Granulation	Coarse irregular	Zn, Sn, Pb, Cu	Low m.p. metals
Reduction	Spongy, porous	Fe, W, Mo, Cu, Ni	Cheap, mass production
Electrolysis	Dendritic, pure	Cu, Fe, Ni, Zn	High purity, costly
Carbonyl	Spherical fine	Fe, Ni	Very pure

3. Characterisation of Metal Powders

* PARTICLE SIZE & DISTRIBUTION: sieve analysis (> 45 micron, mesh no.), sedimentation, microscopy, laser diffraction (very fine).
  PARTICLE SHAPE: spherical, irregular, dendritic, flaky, angular - affects flow and packing.
* APPARENT DENSITY (loose powder mass / volume, Hall flowmeter funnel), TAP DENSITY, FLOW RATE (time for 50 g to pass through a standard funnel, s/50 g), compressibility (green density vs pressure), green strength, sintering activity, chemical purity (hydrogen loss test for oxide content).

Useful relations

$$\text{Porosity } \% = (1 - \rho_{sintered} / \rho_{theoretical}) \times 100$$

Apparent density = mass of loose powder / volume; Flow rate = time (s)

Theoretical density of mixture: $1/\rho = w_1/\rho_1 + w_2/\rho_2$

Eg: Cu-10% Sn bronze (ρ_{Cu} = 8.96, ρ_{Sn} = 7.3 g/cc): 1/rho = 0.9/8.96 + 0.1/7.3 = 0.1141 -> rho = 8.76 g/cc. If sintered density is 7.5 g/cc, porosity = (1 - 7.5/8.76) x 100 = 14.4%.

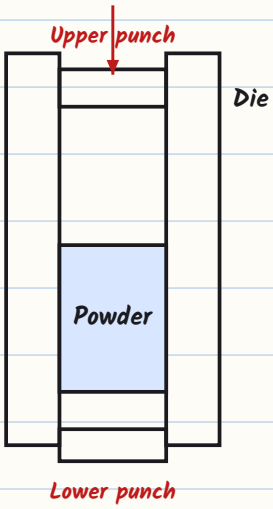
4. Blending and Mixing

* Powders of different metals / sizes are mixed uniformly in V-blender, double cone, drum or Y-mixer for 5 - 30 min. LUBRICANT (0.5 - 1.5% zinc stearate, stearic acid, wax) added to reduce die wall friction and ejection force; binders (for fine powders), additives like graphite for steel.
* Blending = mixing of powders of the same chemical composition (different sizes); Mixing = combining powders of different materials. Over-mixing may cause segregation.

5. Compacting Methods

Aim: to give the loose powder the shape, dimensions, density and 'green strength' (strength to handle before sintering). The pressed part is called GREEN COMPACT.

(a) Conventional die compaction (cold pressing)



Die compaction

Pressure 100 - 700 MPa
Green compact: 70-90% dense, strong enough to handle.
Density gradient due to die wall friction -> double ended pressing / lubricant (zinc stearate).

* Die filled with measured powder (fill depth > compact height, ratio 2:1 to 3:1) -> pressed by upper and lower punches in mechanical, hydraulic or rotary presses -> compact ejected. 100 - 700 MPa, 10 - 300 parts / min.
* Green density about 80-90% of theoretical; pressure raises density; friction gives density variation (corners low). Spring back occurs after ejection (0.1 - 0.3%).

Stages of compaction and effect of pressure

* Stage 1: particles rearrange and fill voids (low pressure). Stage 2: particles deform elastically and plastically, cold weld at contacts, density rises fast. Stage 3: bulk compression, work hardening, density saturates near 90-95% (pores still remain).
* Factors: particle size / shape (irregular gives higher green strength), hardness of powder (soft Cu compacts well, hard W poorly), lubricant, pressing speed, pressure, die design, powder friction (density gradient: use double action pressing or floating die), L/D ratio of part (< 3 recommended).
* Design rules for P/M parts: simple shape that can be ejected from the die; uniform section thickness, avoid thin walls (< 1.5 mm) and sharp

corners, no undercuts or threads perpendicular to pressing direction, generous radii, tapered, minimum hole diameter 2 mm.

(b) Isostatic pressing

* COLD ISOSTATIC PRESSING (CIP): powder in flexible rubber mould in fluid, pressure applied from all directions (200 - 400 MPa) - uniform density, long slender shapes, tubes, large parts.
* HOT ISOSTATIC PRESSING (HIP): pressure 100 - 200 MPa with argon gas at 900 - 1250 C; compaction and sintering together -> near 100% density; super alloy and Ti parts, tool steels, eliminating casting porosity.

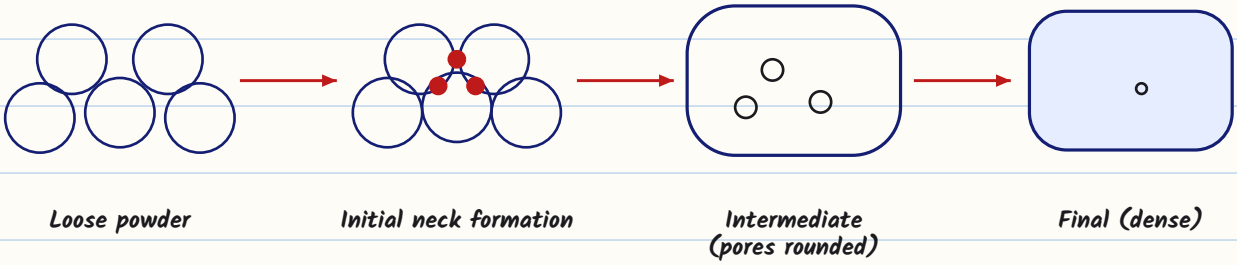
(c) Other compaction methods

* POWDER ROLLING: powder fed between rolls -> continuous strip (Cu, Ni, steel sheet); EXTRUSION: powder in can or with binder extruded; INJECTION MOLDING (MIM): fine powder + 40% polymer binder injected in mould, debinded then sintered - complex small parts (watch cases, medical tools).
* SLIP CASTING (ceramics / metals): slurry poured into plaster mould; HOT PRESSING: pressing and sintering simultaneously in graphite dies (WC, diamond tools); Spark plasma sintering (SPS); forging of sintered preforms (powder forging) ; ADDITIVE MANUFACTURING (laser sintering).

6. Sintering

* SINTERING = heating the green compact in a CONTROLLED ATMOSPHERE at a temperature BELOW the melting point of the main constituent (typically 0.7 - 0.9 T_m; Fe 1120 C, Cu 850 C, Bronze 800 C, WC-Co 1450 C, W 2200 C) for 20-60 min to bond the particles by

atomic diffusion.



Stages of sintering

* 1. Initial stage: particle contact points weld, NECKS form by surface diffusion / vapour transport. 2. Intermediate: necks grow, pores become smooth interconnected tubes, shrinkage and densification (grain boundary and volume diffusion). 3. Final: pores close and become spherical and isolated, slow densification, grain growth.
* Driving force: reduction of total surface energy (surface area) of fine particles. Mechanisms: evaporation-condensation, surface diffusion, volume diffusion, grain boundary diffusion, plastic flow.

Types

* SOLID PHASE sintering: single powder, or mixed powders all remain solid (Fe, Cu, bronze).
* LIQUID PHASE sintering: a lower melting component melts and wets the solid (e.g. WC + Co at 1450 C; Fe + Cu infiltrated; Cu-Sn bronze; W-Ni-Fe) -> faster densification, high density, strong.

Furnace and atmosphere

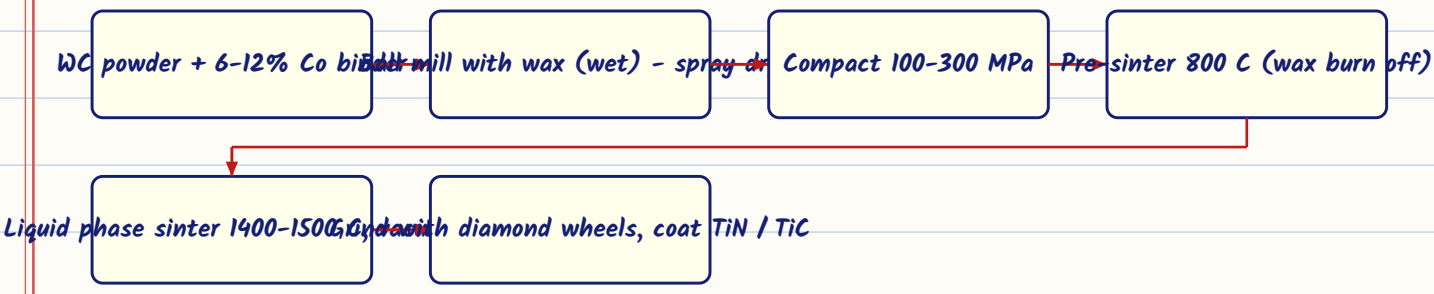
* Continuous mesh belt, walking beam, pusher or batch / vacuum furnaces with 3 zones: pre-heat / burn-off (delubrication at 300-600 C), high heat (sintering), and cooling zones (water jacket, protective atmosphere maintained).

* Atmospheres: endothermic gas (CO, H2, N2), dissociated ammonia (75% H2 + 25% N2), hydrogen, nitrogen-hydrogen, vacuum - to prevent oxidation, reduce surface oxides, control carbon. Never air.
* Variables: temperature, time, atmosphere, heating / cooling rate, green density, particle size. Effects: strength increases (about 2-3 times), density rises, dimensional change (shrinkage or growth 0.1-2%), electrical and thermal conductivity improve, ductility and toughness improve.

7. Methods of Manufacturing Sintered Parts

Route	Steps	Typical parts
Press and sinter (single)	Mix -> press -> sinter	Gears, bushes, cams, structural parts (most common, 70-90% density)
Press - sinter - re-press (re-sizing / coining)	Second pressing improves dimension, density	Close tolerance parts, bearings
Double press - double sinter	Press, pre-sinter, repress, final sinter	High density 92-95%
Infiltration	Sintered porous skeleton filled with molten lower melting metal (Cu into Fe, Ag into W)	Electrical contacts, high strength gears
Impregnation	Pores filled with oil (30 min. vacuum), polymer, wax	Self lubricating bearings
HIP / hot pressing / powder forging	Pressure + temperature	Full density, critical aerospace and tool parts
MIM	Fine powder + binder -> inject -> debind -> sinter	Small complex parts, medical, watch, firearms

Case study 1 : Manufacture of cemented tungsten carbide tools



* Hardness 1400-1800 HV, retains hardness up to 1000 C, used for cutting tool inserts, drills, mining bits, wire drawing dies. Cobalt wets WC during sintering (liquid phase), giving strong, tough compact. Cermet class of material.

Case study 2 : Self lubricating bearings

* Cu (90%) + Sn (10%) bronze powder + graphite 1%, press to 6.4-6.8 g/cc, sinter at 800 C, size, then vacuum impregnate with oil (about 20% by volume). Oil is drawn to the surface by heat of friction and returns by capillary action. Used in fans, domestic appliances, automotive starters, wipers.

Case study 3 : Friction materials, magnets, contacts

* Brake and clutch linings: Cu / Fe matrix + friction modifiers (silica, graphite). Alnico, ferrite and Nd-Fe-B magnets pressed in a magnetic field and sintered. Electrical contacts: Ag-W, Cu-W, Ag-CdO: pressed, sintered and infiltrated (good arc erosion, high conductivity).

8. Secondary Operations

* SIZING / COINING (re-pressing): improve dimensional accuracy and density; MACHINING (drilling, turning, tapping) - limited by pores;

carbide tools used.

* IMPREGNATION with oil (self lubricating bearings hold 10-30% oil), resin (leak sealing, pressure tightness) ; INFILTRATION with Cu; HEAT TREATMENT: steam treatment (blue oxide film Fe3O4 for wear and corrosion), carburising / hardening, case hardening, tempering.
* PLATING and coating (Cu, Ni, Zn, Cr), deburring / tumbling, grinding, welding / joining (sinter bonding), machining, polishing, forging.

9. Applications of P/M Products

* Self lubricating (oil-impregnated) bronze and iron bearings (Oilite) - bushes for home appliances, motors, auto; porous filters (bronze, stainless, Ni, Ti) for fuel, oil, chemical and air.
* Cemented carbide cutting tools (WC + 6-12% Co), dies, mining and drilling bits, wire drawing dies; diamond impregnated tools; ceramic cutting tool inserts.
* Automobile: gears, connecting rods, cams, valve guides, valve seats, synchro hubs, oil pump gears, ABS sensor rings, clutch plates, brake pads (friction materials), piston rings; electrical contacts (W-Cu, Ag-W, Ag-CdO), brushes (Cu-graphite) and relays.
* Refractory metals: tungsten filaments, Mo, Ta capacitor anodes, rocket nozzle parts; Magnets: Alnico, ferrites, rare-earth (Nd-Fe-B, Sm-Co) magnets; heavy alloys (W-Ni-Fe) for counterweights and radiation shields; nuclear fuel, aerospace super alloy discs; surgical implants (porous Ti); powder metal gears for tools; filters.

Advantages and limitations

Advantages	Limitations
Near net shape, minimum waste (97% material use)	Costly metal powders and tooling (dies)
Close tolerances, good finish, no machining	Size limited (press capacity, usually < 2.5 kg)
Controlled porosity, unique combinations	Lower strength / toughness than wrought due to porosity
Suited to very high melting point metals	Complex shapes (undercuts, re-entrant) difficult
Mass production, low labour, no pollution	Economic only for large production volume (> 10,000)

Comparison with casting

Powder metallurgy	Casting
Very high melting metals possible	Limited by furnace / mould capability
Solid state process; good uniformity	Liquid state; segregation, shrinkage, gas defects
Porosity 5-25% (controlled)	Usually dense (pores are defects)
High tooling cost - for mass production	Moulds cheap - good for single / big parts
Accuracy +- 0.025 mm	Accuracy +- 0.25 mm and more

Important questions

* 1. Define powder metallurgy; explain the steps with a block diagram; advantages and limitations.
* 2. Explain atomisation, reduction of oxides and electrolytic deposition methods of powder production.
* 3. Describe milling in a ball mill; critical speed of the mill.

* 4. What is sintering? Stages, atmosphere, variables, types of sintering.
* 5. Write short notes on isostatic pressing, infiltration, impregnation, MIM.
* 6. List applications of P/M parts with suitable examples.

Revision: P/M steps - powder production, blending, compaction, sintering, finishing. Powder methods - milling, atomization, granulation, reduction, electrolysis. Sintering temp = 0.7 - 0.9 Tm in protective atmosphere.